
Data Analysis

Course Description: Understanding statistics is vitally important for work in Psychology. Data analytic techniques form the bedrock for many avenues of scientific research, and understanding their theory implies knowing when a particular approach is appropriate and, more critically, when it is not. Did your clinical intervention work? Is the hot new teaching method effective? Can we predict what someone will do under duress? Is eyewitness testimony accurate? In each of these scenarios, psychology and statistics intertwine to provide narrative and are used to sway opinion, serve as evidence, and communicate our findings to a broad audience. In this course: Introduction to Data Analysis in Psychology, you will gain the foundation that other courses in your degree build upon that will enable you to conduct research, interpret results, and grasp the more complex statistical models that are widely used in our field.

Required Text: Privitera, G. (2018). *Essential Statistics for the Behavioral Sciences* (2nd Ed.). Sage Publishing. Student study guide also available (not required) and online resources:

- The course focuses on conceptual understanding over computation – study accordingly!
- Further, you will contribute to your peers' understanding of the material. You are a key component to our classes' success and your participation, especially during tutorial, is expected!
- **Calculator:** A calculator is required for this course and only basic or scientific (non-graphing) calculators are allowed in the exam.
- The course is designed to optimize learning by requiring active participation (through the tutorials) and distributed practice (exercise questions, worksheets, assignments, and exams). Remember learning this material is a cumulative process, so if you find yourself falling behind, please let me or your TA know as soon as possible. We are here to help you learn the material, so use us!

COURSE EVALUATION

- **Assignment 1** (submit by 11:59PM on October 9th): 10%
- **Midterm Exam** (held during class on November 2nd): 30%
- **Assignment 2** (submit by 11:59PM on November 27th): 15%
- **Final Exam** (held in-person during the exam period, date TBA): 35%
- **Tutorial Citizenship:** 10%

Assignments: You are encouraged to complete both assignments using pen and paper and scan or take pictures of your work. Your final submissions should be submitted as **a single PDF file** by their respective deadline. If a technology issue arises at the time of submission, please email a copy of your assignment to your TA. For these assignments, all work is assumed to have been done independently. While we encourage group work or collaboration with other students for the purposes of studying and mastering the material, this is **not allowed** for evaluative components and may be considered academic dishonesty.

Exams: The exams will cover material presented in the lectures and textbook readings. Questions will be constructed using a combination of **conceptual** and **computational** multiple-choice and short answer items. While a calculator is required, please note cellphones and graphing calculators are not acceptable and will be confiscated for the duration of the exam. The final exam is cumulative in the sense that the post-midterm content builds upon the previous content, but focus will be on post-midterm material.

Tutorial Citizenship: Learning statistics is an active process. This component of your grade reflects your commitment to your learning and your contribution to your peers' success. Your TA will be assessing your attendance, participation, and overall dedication to the tutorial environment.

Final Grades: The range of values listed below reflect anticipated values for letter assignments only; the actual values may vary depending on the performance of the class, will be based upon unrounded averages, and *final grades may be scaled if necessary*.

A-/A/A+ = 85+; B-/B/B+ = 75-84; C-/C/C+ = 65-74; D = 50-64; F = 0-49.

Tentative Course Schedule (Date/Topic/Reading/Tutorial)

2026/09/14	Course Introduction and Basic Concepts	Chapter 1	Y
2026/09/21	Data Types and Visualization	Chapters 2 and 3	Y
2026/09/28	Measures of Central Tendency and Variability	Chapter 4	Y
2026/10/05	Standard Scores	Chapter 5	Y
2026/10/09	Assignment 1 Due (10%)	--	--
2026/10/12	Thanksgiving (NO CLASS/LABS)	--	N
2026/10/19	Probability and the Central Limit Theorem	Chapter 6	Y
2026/10/26	Hypothesis Testing	Chapter 7	Y
2026/11/02	MIDTERM (30%)	--	N
2026/11/09	t-tests (One-sample tests; Ind. measures)	Chapters 8 and 9	Y
2026/11/16	t-tests (Dep. Measures) and ANOVA	Chapters 10 and 11	Y
2026/11/23	Introduction to Correlation	Chapter 13	Y
2026/11/27	Assignment 2 Due (15%)	--	--
2026/11/30	Introduction to Regression	Chapter 13	Y
2026/12/07	Categorical Data Analysis/Course Wrap-Up	Chapter 14	Y
2026/12/09-20	FINAL EXAM (35%; cumulative)	--	N

Supplementary (Optional) Resources:

- Lane, D.M. et al. (n.d.). *OnlineStatBook* (<http://onlinestatbook.com>). This is a free resource that can be accessed online in a variety of formats and features interactive applets and videos.
 - Gravetter, F.G., & Wallnau, L.B., Forzano, L.B., & Witnauer, J.E. (2021). *Essentials of Statistics for the Behavioral Sciences* (10th ed.). Cengage.
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